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NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT
2026
Even Semester - 2025-26
IT201 - DATA STRUCTURE AND ALGORITHMS

Time: 3 Hrs

Maximum Marks : 60

Instructions to the candidate:

Figures to the right indicate full marks.

Draw neat sketches and diagram wherever is necessary.

Candidates are required to give their answers in their own words as far as practicable

Part A

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1. The traversal of a BST that outputs keys in ascending order is _____. (1) C03 BL3 PO4
(a) preorder (b) postorder
(c) inorder (d) level order
2. If an algorithm finishes after a fixed number of steps regardless of input size, its time complexity is usually _____. (1) C01 BL1 PO3
(a) $O(1)$ (b) $O(n)$
(c) $O(n \log n)$ (d) $O(n^2)$
3. The load factor of a hash table is defined as _____. (1) C04 BL4 PO4
(a) number of collisions / table size (b) number of items / table size
(c) table size / number of items (d) number of items / collisions
4. Choose the correct postfix expression of the given infix expression $A + (B * C / D + (E + F) / G)$. (1) C02 BL3 PO3
(a) $ABC * D / EF + G / ++$ (b) $ABCD / * EF + G / ++$
(c) $ABC * D / E + FG / ++$ (d) None of these
5. A stack follows the _____ order. (1) C02 BL2 PO3
(a) FIFO (b) LILO
(c) LIFO (d) random
6. A threaded binary tree is designed mainly to _____. (1) C03 BL4 PO4

- (a) increase the height of the tree (b) store duplicate keys only
(c) remove all leaf nodes (d) replace NULL links with useful links
7. The **B Tree** is primarily designed to optimize _____. (1) C03 BL4 PO4
(a) recursive function execution (b) secondary storage access operations
(c) arithmetic expression evaluation (d) sequential memory traversal
8. Prim's algorithm is used to find a _____. (1) C04 BL4 PO4
(a) shortest path tree (b) topological order
(c) binary search tree (d) minimum spanning tree
9. In an expression tree, leaf nodes represent _____. (1) C03 BL4 PO4
(a) operands (b) operators
(c) parent pointers (d) None of these
10. Which of the following data structures are primarily used in Breadth First Search (BFS) and Depth First Search (DFS) respectively? (1) C03 BL2 PO2
(a) Stack, Queue (b) Stack, Linked List
(c) Array, Stack (d) Queue, Stack
11. Interpolation search is most effective when the data are _____. (1) C04 BL4 PO4
(a) uniformly distributed (b) reverse sorted
(c) randomly distributed (d) arranged in linked structure
12. Which operation is not directly supported in $O(1)$ time by an array? (1) C02 BL3 PO3
(a) Random access by index (b) Reading an element
(c) Insertion in the middle (d) Updating an element

Part B

(Answer any four of the following) 5x4=20

13. Write a recursive **function/algorithm** to solve the **Tower of Hanoi** problem for n disks using three rods SOURCE, AUXILIARY, and DESTINATION. (5) C02 BL4 PO3
14. Consider a Circular Queue of size 7 implemented using an array. Initially, FRONT = 3, REAR = 5. (5) C02 BL4 PO3

Index	0	1	2	3	4	5	6
Queue	—	—	—	S	J	B	—

Perform the following operations sequentially and show the status of the Circular Queue (FRONT, REAR, and queue contents) after each operation:

- i) Insert E
- ii) Insert R
- iii) Delete two elements
- iv) Insert S, T
- v) Delete one element
- vi) Insert A
- vii) Insert C

Also indicate clearly if any Overflow or Underflow condition occurs during the operations.

15. Consider a postfix expression consisting of operands and arithmetic operators (+, −, *, /). **Develop the algorithm for evaluating** a postfix expression using the STACK data structure. (5) C02 BL3 PO3
16. Consider a Singly Linked List in which only one pointer HEAD is maintained, pointing to the first node of the list. Write the algorithm to insert a new node containing an element ITEM at the end of the linked list. Also, mention the condition when the linked list is initially empty. (5) C02 BL3 PO3
17. Define asymptotic notation and explain each type. (5) C01 BL2 PO1
18. Construct the Binary Tree from the following traversal sequences. Show all the steps . (5) C03 BL4 PO2

Inorder Traversal:

$a \rightarrow b \rightarrow c \rightarrow d \rightarrow e \rightarrow f \rightarrow g \rightarrow h \rightarrow i \rightarrow j \rightarrow k$

Preorder Traversal:

$f \rightarrow b \rightarrow a \rightarrow d \rightarrow c \rightarrow e \rightarrow i \rightarrow h \rightarrow g \rightarrow k \rightarrow j$

Part C

(Answer any three of the following) 3x10=30

19. (a) Develop the algorithm for the Merge Sort technique using recursion. (4) C04 BL4 PO3
- (b) Illustrate the Merge sorting process for the following list using suitable steps:
38, 43, 27, 3, 2, 82, 10 (2) C04 BL4 PO3
- (c) Determine the time complexity of Merge Sort in the best case. (4) C04 BL4 PO3
20. (a) A two-dimensional array Arr[15][25] is stored in contiguous memory locations using **Column Major Order**. Each element of the array requires 4 bytes of storage. The base address of the array, i.e., the address of Arr[0][0], is 4000.

Calculate the memory address of Arr[12][9]. (3) C02 BL3 PO3
- (b) Construct the corresponding Expression Tree for the following arithmetic expression.
Clearly show the hierarchy of operators (within circles) and operands (within rectangles) in the tree.
- $$\left(\frac{A + B \times C}{D^2 - E} \right) \times F + G$$
- (c) Define a Priority Queue data structure. (4) C03 BL2 PO2
Also discuss the various implementation/design approaches that can be used to realize a Priority Queue.
[1+3 Marks]
21. (a) Insert the following keys into an initially empty AVL Tree in the given order: (5) C01 BL1 PO1
40, 20, 10, 25, 30, 22, 60, 70, 65, 80, 75, 55

Construct the AVL Tree after each insertion and clearly indicate all balancing operations/rotations (LL, RR, LR, and RL) performed during the construction process.
- (b) Develop the non-recursive (iterative) algorithm/function for performing Binary Search on a sorted array. (5) C04 BL3 PO3
Assume that the array elements are stored in ascending order. The algorithm should search for a given element ITEM and report whether the element is found or not.

22.

- (a) Construct a B-Tree of order 5 by inserting the following keys in the given sequence:

(6) C03 BL4 PO3

12, 8, 2, 25, 6, 14, 28, 17, 7, 52, 16, 48, 68, 3, 26, 29, 53, 55, 45

Show the structure of the B-Tree after each insertion.
Clearly indicate all node splits, key promotions, and changes in the tree structure during the insertion process.

- (b) Consider a Doubly Linked List in which each node contains three fields: PREV, DATA, and NEXT. The pointer HEAD points to the first node of the list, and a pointer DHEAD points to a particular node in the doubly linked list.

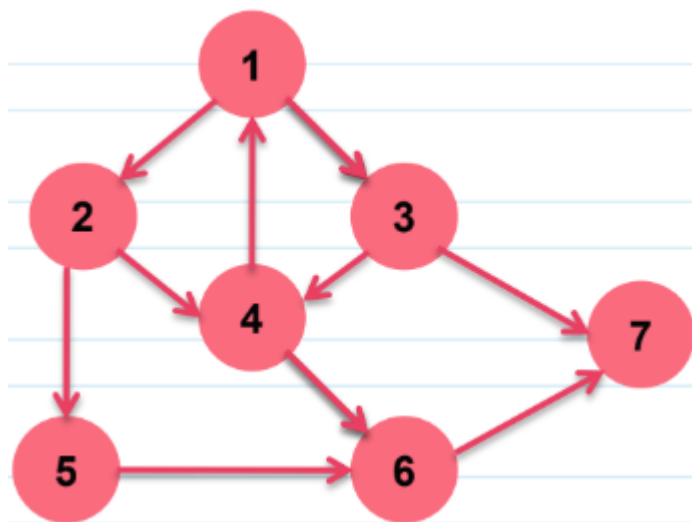
(4) C02 BL3 PO3

Write the algorithm/function to insert a new node containing the element ITEM immediately after the node pointed to by DHEAD.

Also update all necessary links (PREV and NEXT) after insertion.

23. (a) Do the BFS Traversal of the graph.

(5) C03 BL4 PO1



- (b) Convert the following infix expression into equivalent reverse polish notation expression using stack. $(A+B)*C-(D-E)/(F+G)$

(5) C02 BL4 PO1